

AI4S交叉实践课程

----- 计算认知科学实践：从心智机制到神经网络模型

弭元元

清华大学 心理与认知科学系

助教： 林长涵 宁睿轩 冯至韬 邵逸峰 姚立昭

About Myself

教育背景:

物理学教育背景

- 物理学 学士，2007
- 理论物理 博士，2012

工作经历:

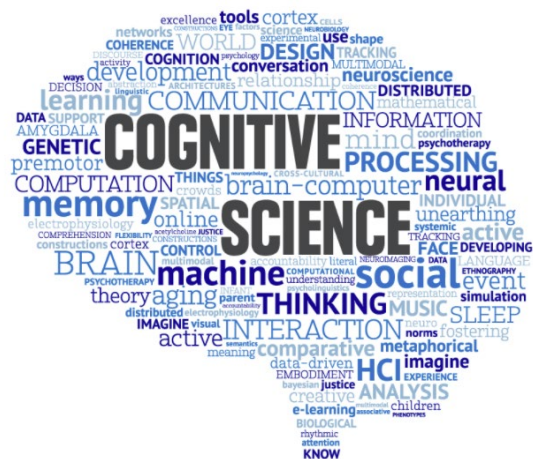
计算神经科学

- 2012-2014: 苏州大学 系统生物学研究中心，讲师
北京师范大学，日本理化学研究所（RIKEN），访问学者
- 2014-2016: Weizmann Institute of Science, Israel
Columbia Univ., US, 博士后
- 2016-2018: 军事医学研究院 军事认知与脑科学研究所，研究员
- 2018-2023: 重庆大学 医学院 神经智能研究中心，长聘教授
- 2023- : 清华大学 心理系，长聘副教授

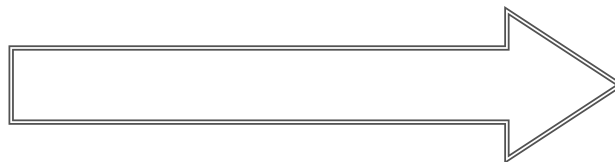
计算认知神经科学
与类脑计算

A Bridge from Brain Cognitive Science to AI

为类脑智能提供新思想和新模型

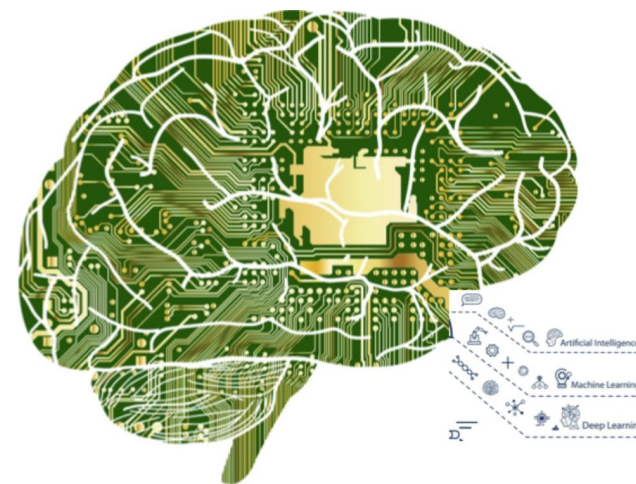


脑认知、心理



桥梁：

计算认知神经科学



类脑智能

What is Computational Cognitive Neuroscience

- 1907: LIF model 脉冲神经网络、类脑芯片
- 1950s: HH model (Hodgkin & Huxley)
- 1960s: Roll's cable equation
- 1970s: Amari, Wilson, Cowan et al.
- 1982: Hopfield model (Amari-Hopfield model)
- 1988: Sejnowski et al. "Computational Neuroscience"

Koch和Bower首次在Woods Hole的海洋生物学所开办summer school

The ultimate aim of computational neuroscience is to explain how electrical and chemical signals are used in the brain to represent and process information. This goal is not new, but much has changed in the last decade. More is known now about the brain because of advances in neuroscience, more computing power is available for performing realistic simulations of neural systems, and new insights are available from the study of simplifying models of large networks of neurons. Brain models are being used to connect the microscopic level accessible by molecular and cellular techniques with the systems level accessible by the study of behavior.

What is Computational Cognitive Neuroscience

- Highly multidisciplinary, related to biology, physics, mathematics, computer science, engineering, and AI
- Two aims:
 - Using mathematical models/approaches to elucidate brain functions
 - Developing brain-like computational algorithms
- Challenges: data + math; infant of physics, theoretical physics

对应于物理学的第谷-伽利略时代，大脑工作原理还缺乏清晰的理论！

Mission of Computational Cognitive Neuroscience

What I can not create, I do not understand

--Richard Phillips Feynman

“只有当你试图去制造一个大脑时，你才能真正体会到大脑的奥妙”

----Steven Pinker

What I can not build a computational model, I do not understand

--an anonymous computational neuroscientist

什么是智能？

□ 智能的发生：

- 皮亚杰：智能就是当你不知道怎么办时，调用的东西

□ 智能的体现：

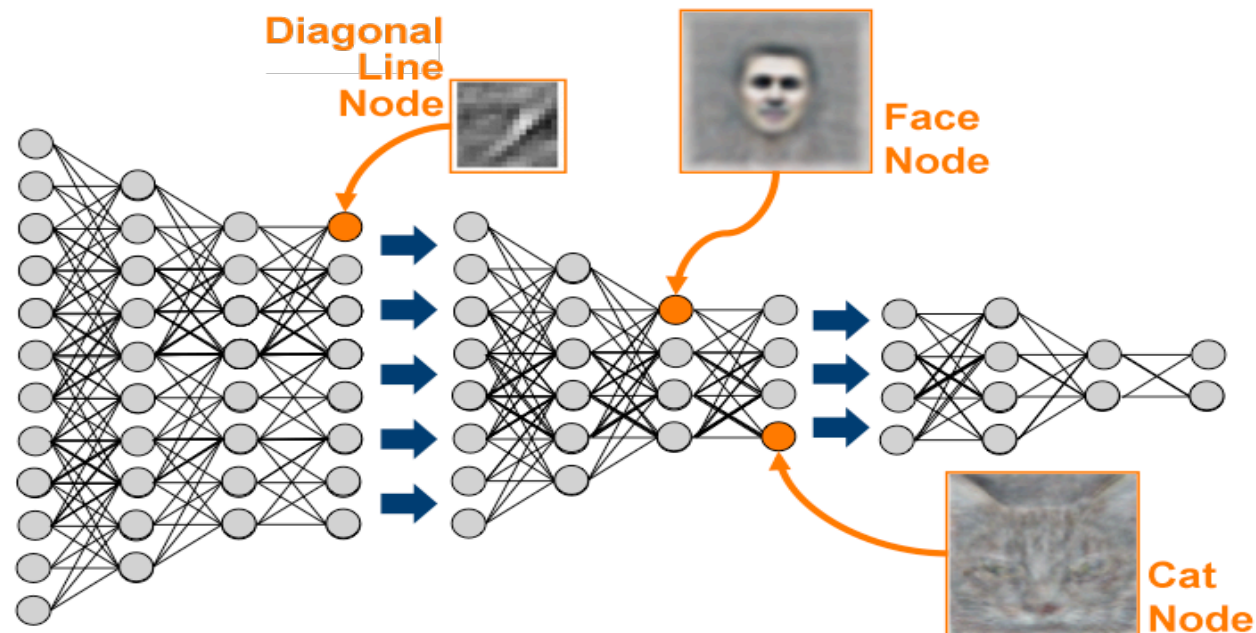
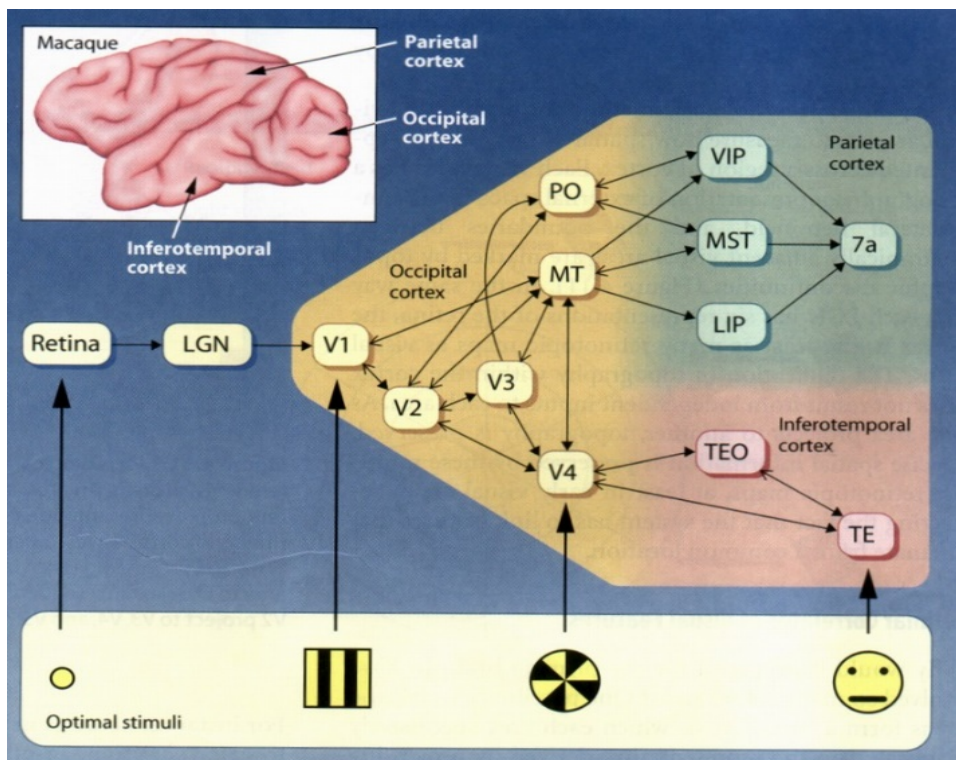
- Hawkins（掌上电脑Palm）、郭爱克：智能就是感知-预测-抉择过程

□ 智能的结果：

- 智能就是人类不断把一个个很“智能”的事情变成不再那么“智能”的行为活动

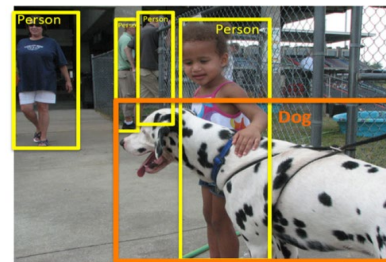
从脑认知中学什么？

□ 深度学习



IMAGENET Large Scale Visual
Recognition Challenge (ILSVRC) 2010-2014

20 object classes — 22,591 images
200 object classes 456,567 images DET
1000 object classes 1,431,167 images CLS-LOC



<http://image-net.org/challenges/LSVRC/>

深度学习的成功



AlphaGo的核心技术:

深度学习算法、强化学习算法、和蒙特卡洛搜索算法

AlphaGo的成功要素:

- 大数据训练方法和高性能计算:

围棋的难度在于其状态空间“无限复杂”，很难用穷举的搜索方法来寻找围棋的优化下法

- AlphaGo设计师Hassabis的多学科交叉背景:

熟悉国际象棋；剑桥的数学/计算机背景；Gatsby计算神经科学中心取得博士学位；MIT机器学习算法

AlphaFold: 蛋白质折叠

深度学习的局限

➤ 对抗样本:



熊猫

+ .007 ×



=

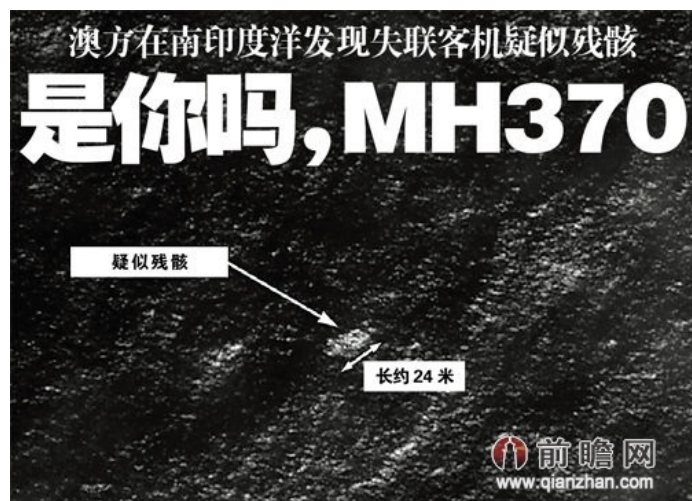


长臂猿

➤ 小样本学习

➤ 举一反三

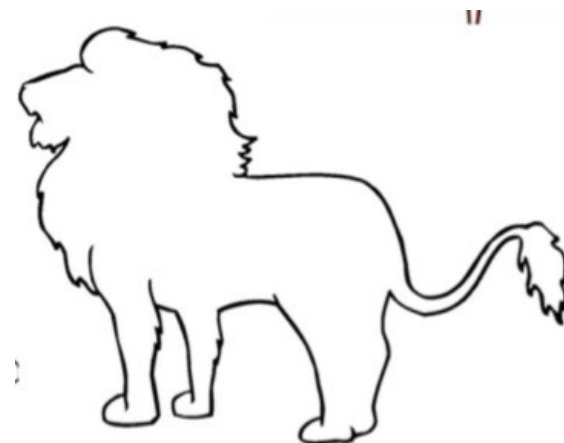
➤ 图像理解



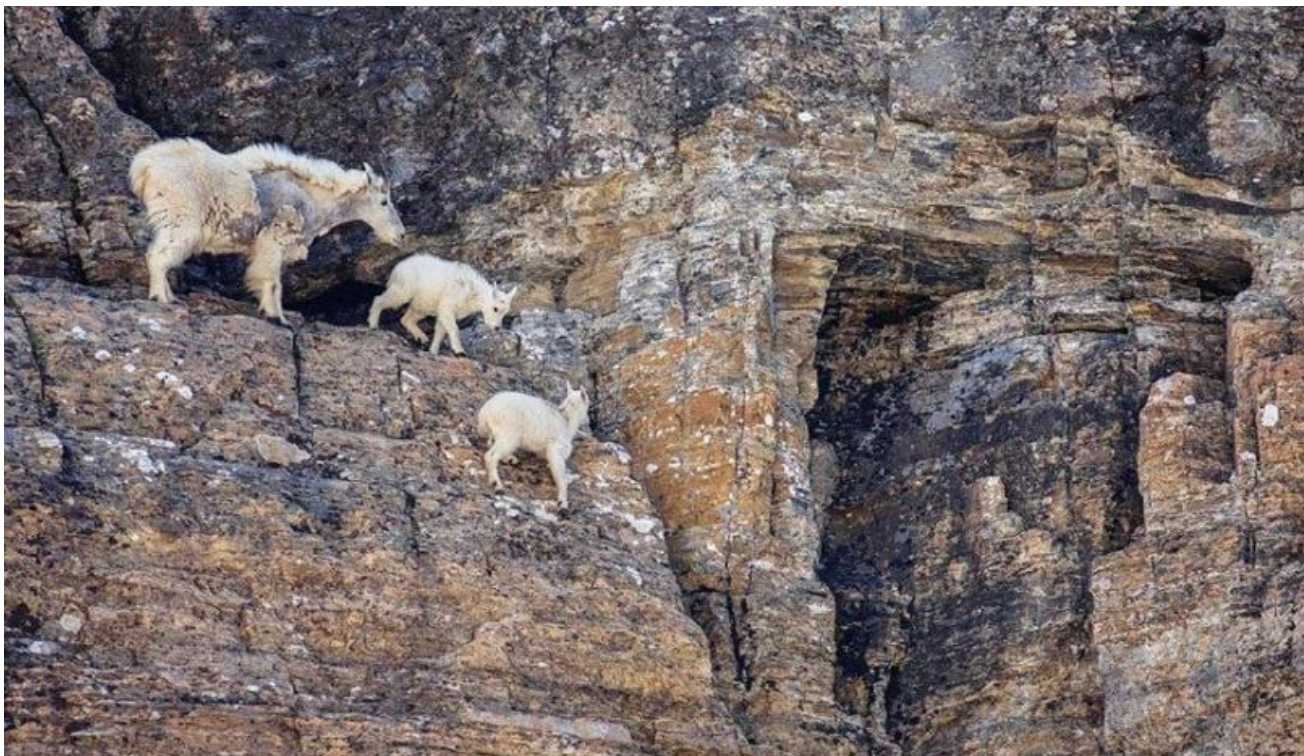
➤ 全局认知



➤ 运动识别



We are far from understanding Intelligence



We are far from understanding Intelligence

波士顿动力公司
Spot机器狗35年进化史

机器人未来能做什么？
波士顿动力公司人形机器人进化史
10年演变攻克运动平衡难题惊骇世俗

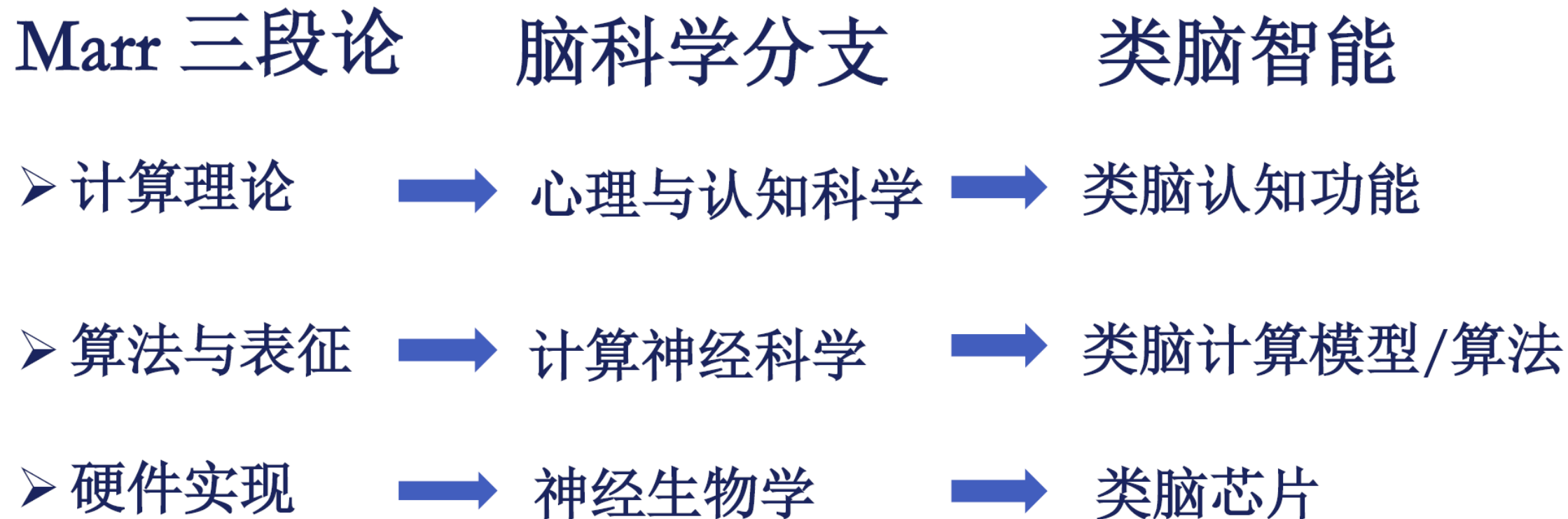
- **莫拉维克悖论**：对机器人而言实现逻辑推理等人类高级智慧只需要相对很少的计算能力，而实现感知、运动等智慧行为却需要巨大的计算能力。

1986年

2009年

Boston Dynamics

十年前的科幻
如今变成了现实



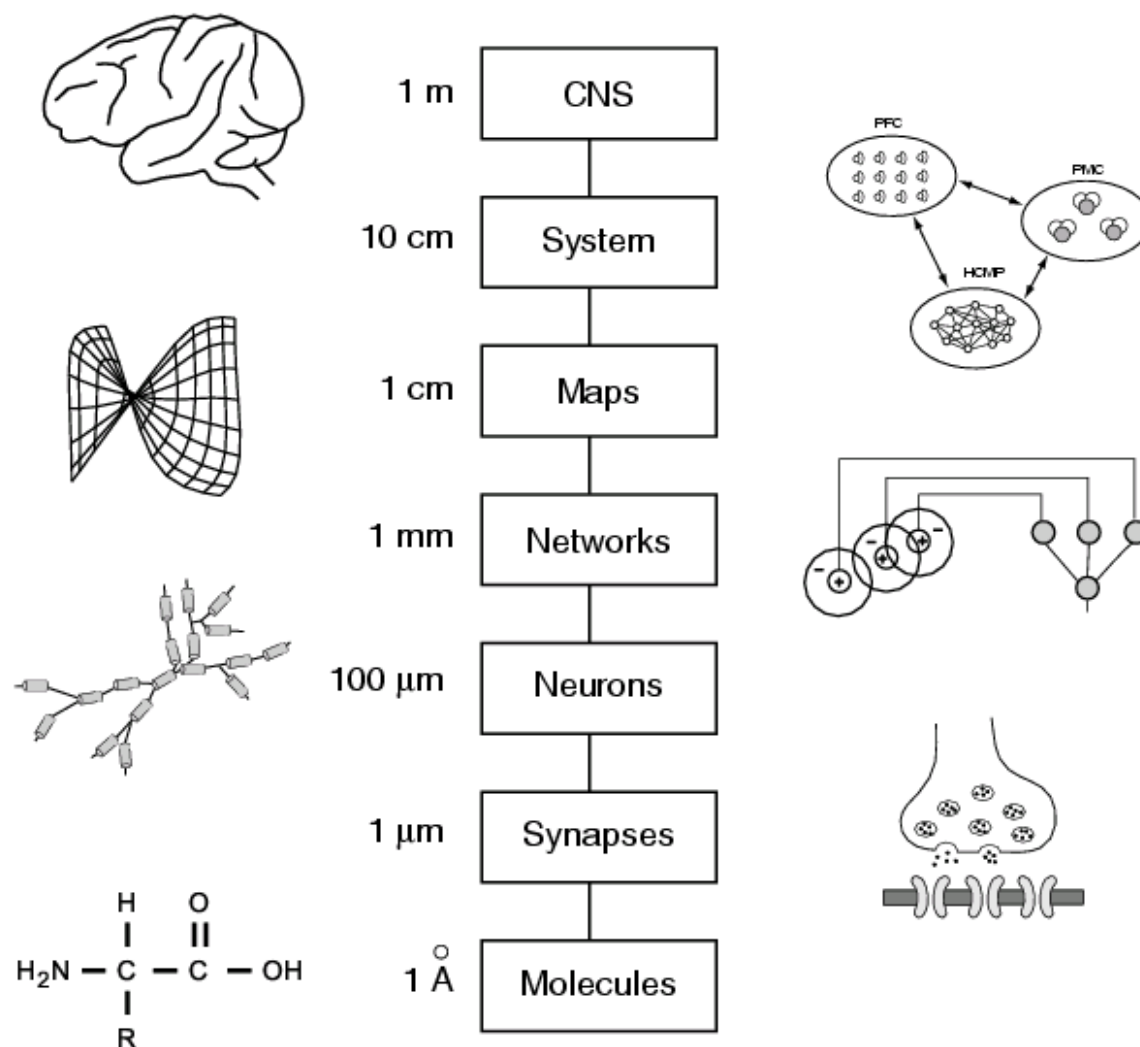
The Computational Neural Representation

The brain is extremely complex: around 10^{11} neurons, each on average with 10^3 connections

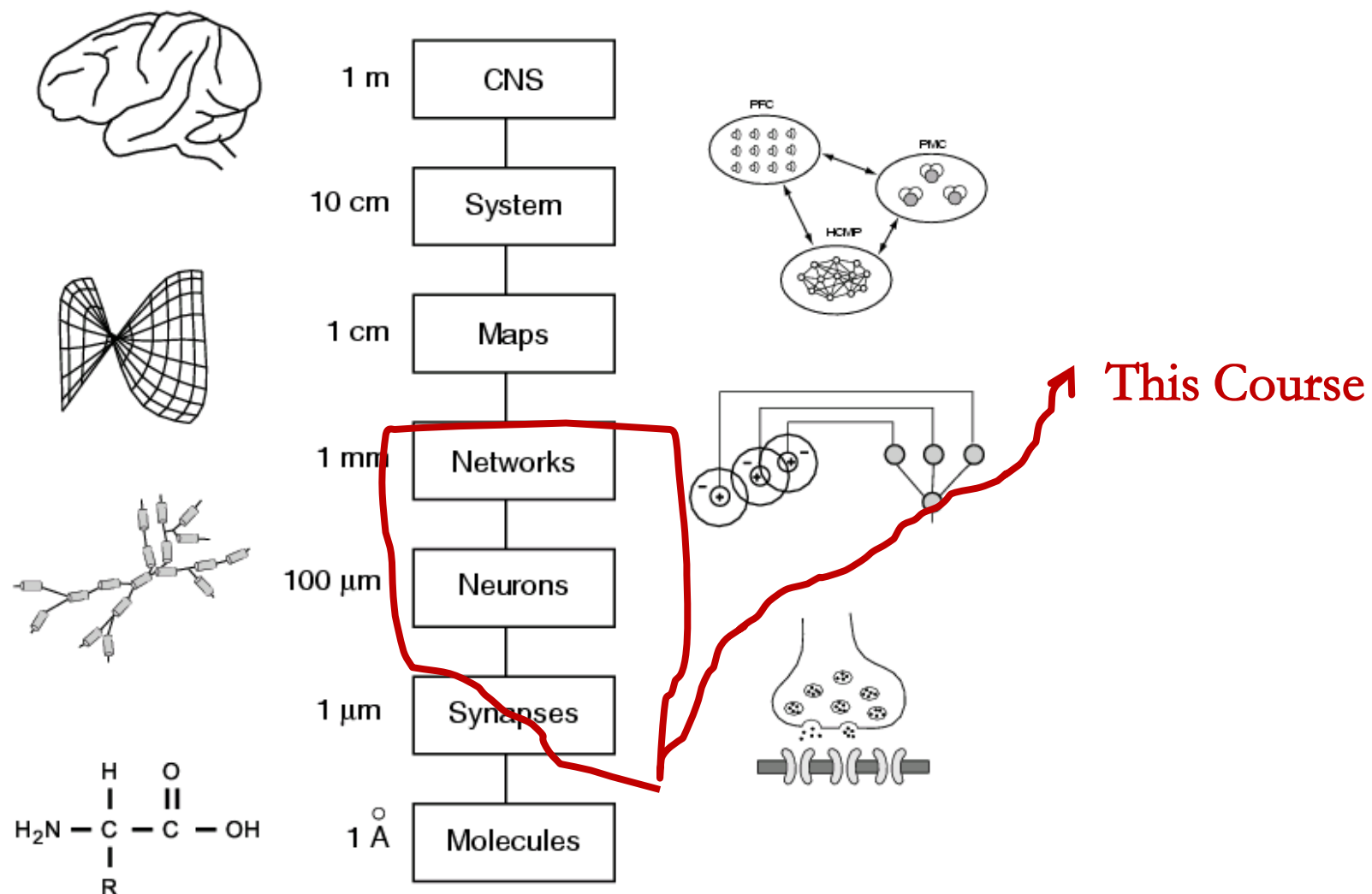
- **Neuron:** the computational unit
- **Synapse:** neurons are connected
- **Network:** neurons are organized into layers

Hypothesis (connectionism):

It is the networks formed by large number of neurons that carry out brain functions!



Structure Levels of Neural Systems



Shaping Sequence Attractor Schema

in Recurrent Neural Networks and its Applications

The Spatiotemporal Information Processing

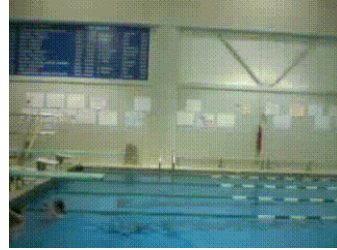
- ✓ Spatiotemporal pattern recognition is fundamental to brain cognitions.



Play basketball



Biking



Diving



Golfing

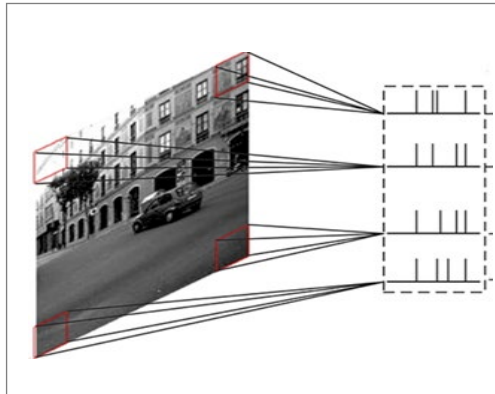


Horse riding

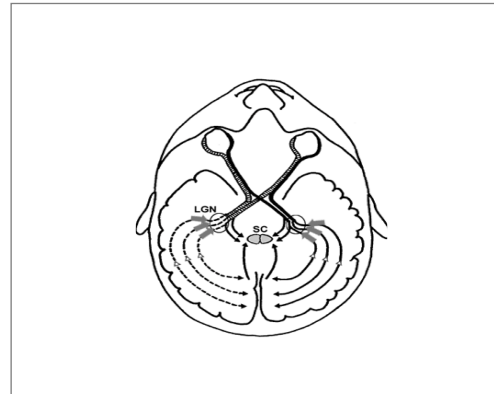
- ✓ The brain is actually tasked with the problem of spatiotemporal pattern recognition, rather than static image classification.



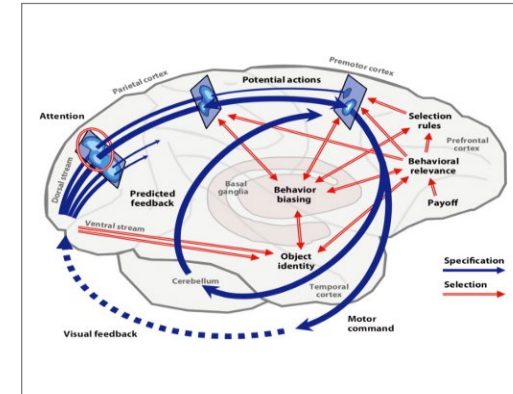
Visual Optical Flow



Spiking Train Propagation



Information Processing



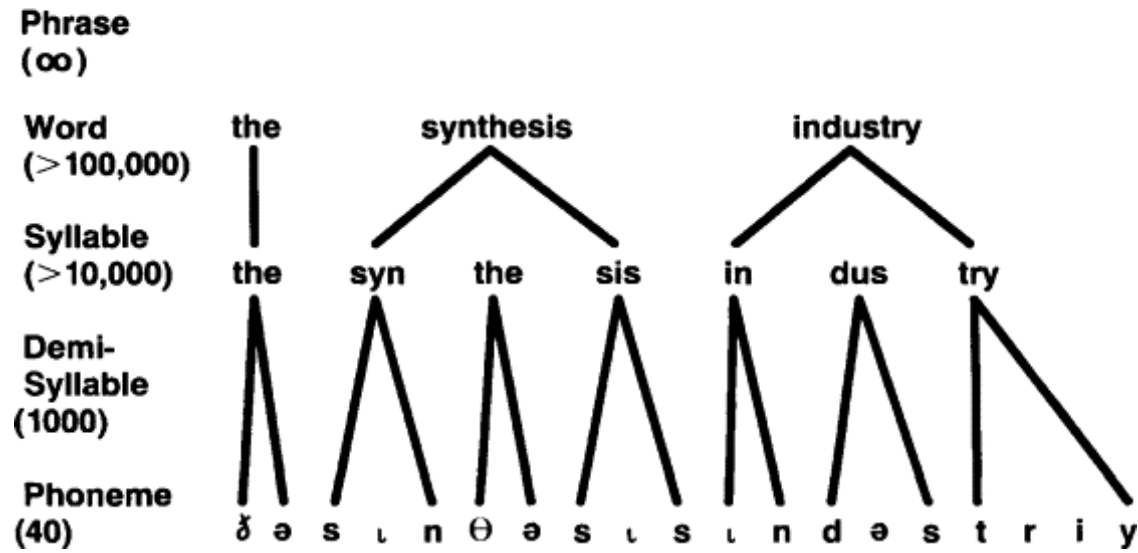
Information Integration

The Spatio-temporal Sequence

- ✓ A spatio-temporal sequence involves a set of **items or events (contents)** that unfold in a specific **order** over time.

Temporal sequence processing is fundamental in many cognitive functions, such as speech recognition, language comprehension, motor control, and memory formation.

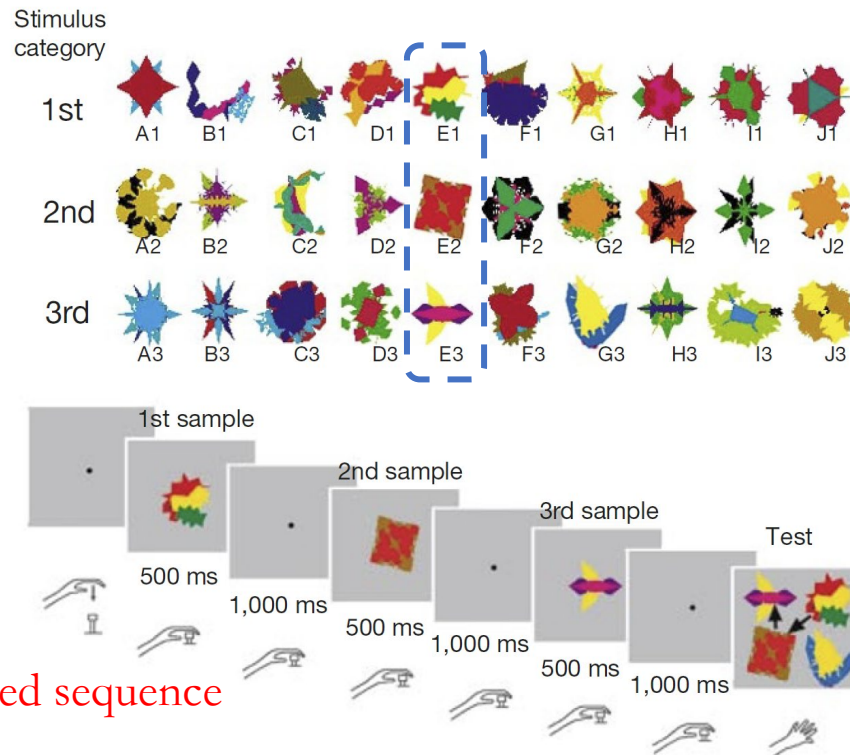
Choosing A Basic Sound Unit



How to Process the Temporal Sequence

Whether a list of arbitrary items is remembered by **associations between sequential items** or by **associations between each item and its ordinal position** ?

Orlov et al, *Nature*, 2000



The delayed sequence recall task

Ten sequences with 3 items:

a1-a2-a3

b1-b2-b3

The most common error : touching the distractor when it shared the same ordinal position as the correct image

e1-e2-**g3**

not e1-**g3**-e3

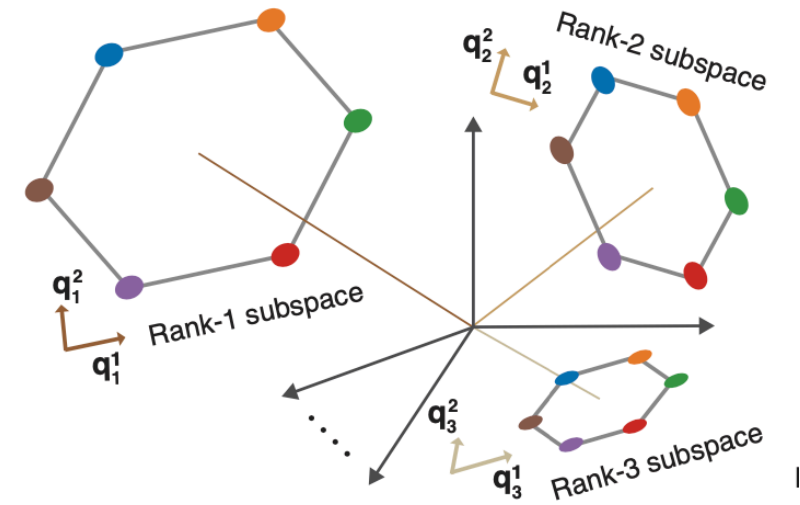
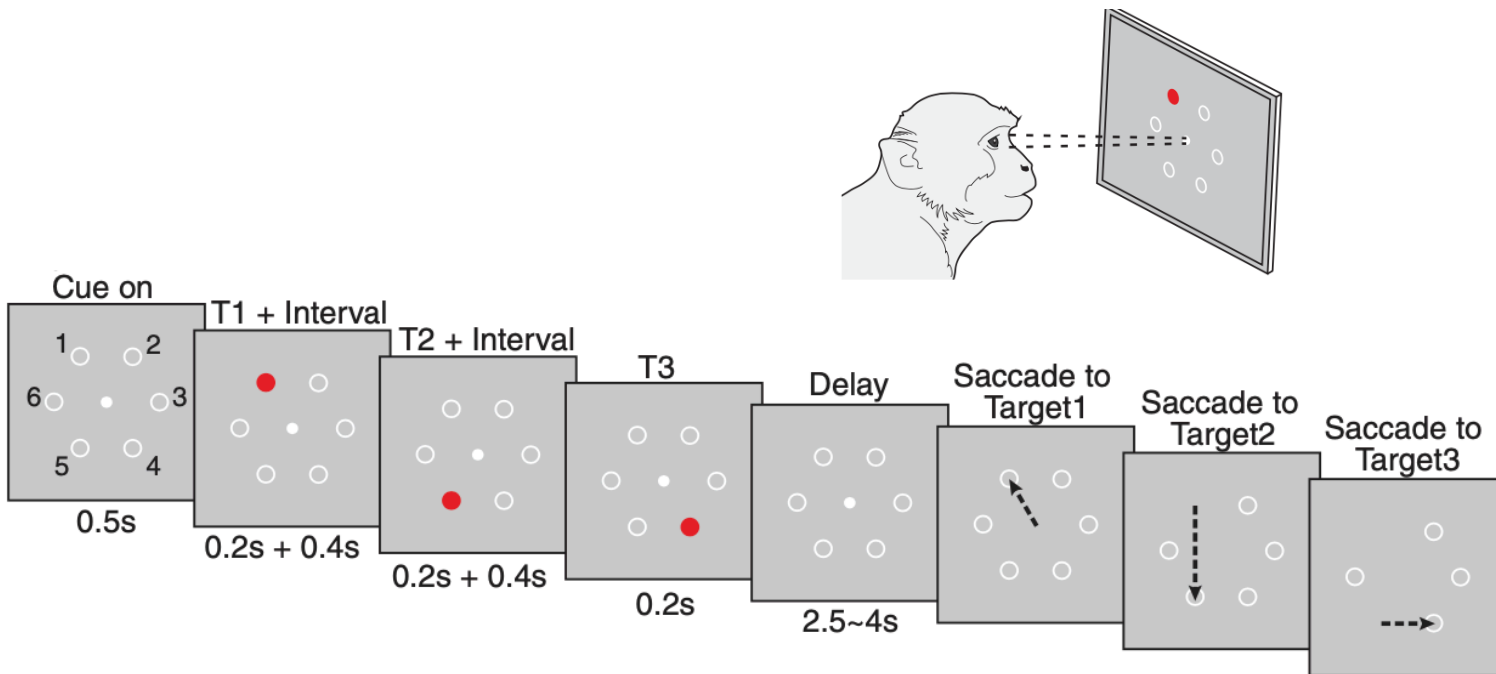
Macaque monkeys categorize images **by their ordinal number**

- ✓ Effective temporal sequence processing relies on **extracting the temporal structure of a sequence**, which includes particularly the ordinal information, i.e., the order of events' occurrence.

The Disentangled Representation of Contents and Ordinal Information

- ✓ A large volume of studies has suggested that the representations of **ordinal information** and **contents** of a temporal sequence are **disentangled** in the brain.
- Studies using single-unit recording in primates have found that a significant proportion of neurons in the dorsolateral prefrontal and intraparietal cortexes are sensitive to the temporal order of presented visual items, but not to their physical attributes.

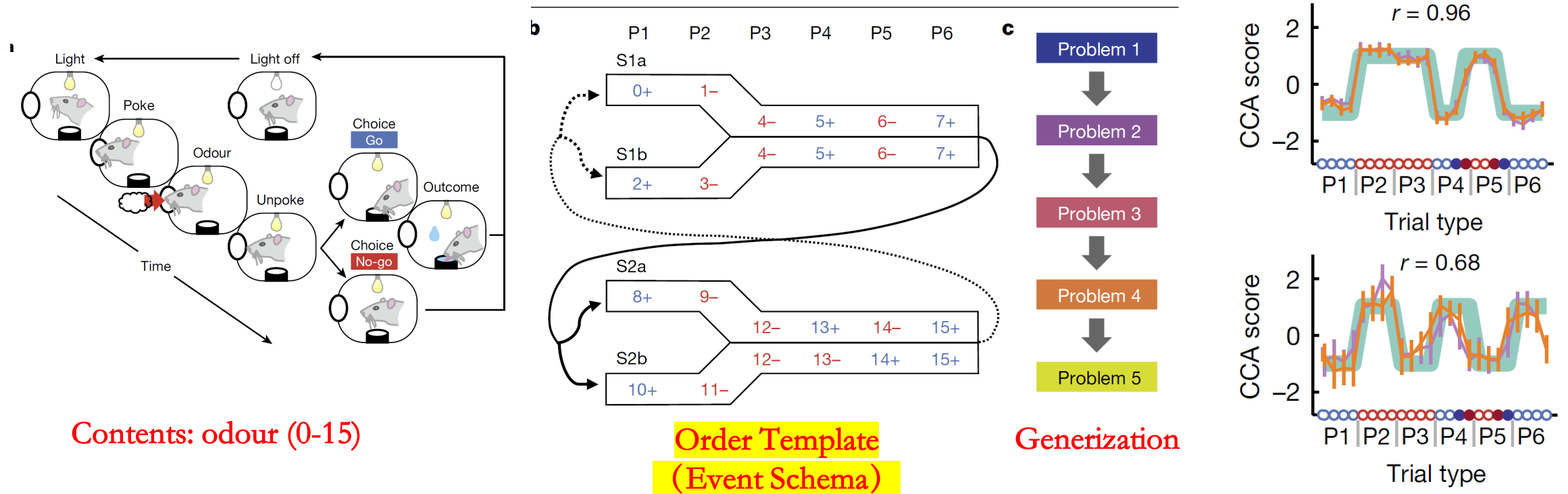
--- Gwilliams et al, *Nature Commun.*, 2022; Xie et al, *Science*, 2022; Yi et al, *Neuron*, 2022; Ninokura et al, *JNP*, 2004



--- Gwilliams et al, *Nature Commun.*, 2022; Xie et al, *Science*, 2022; Yi et al, *Neuron*, 2022; Ninokura et al, *JNP*, 2004

The Disentangled Representation of Contents and Ordinal Information

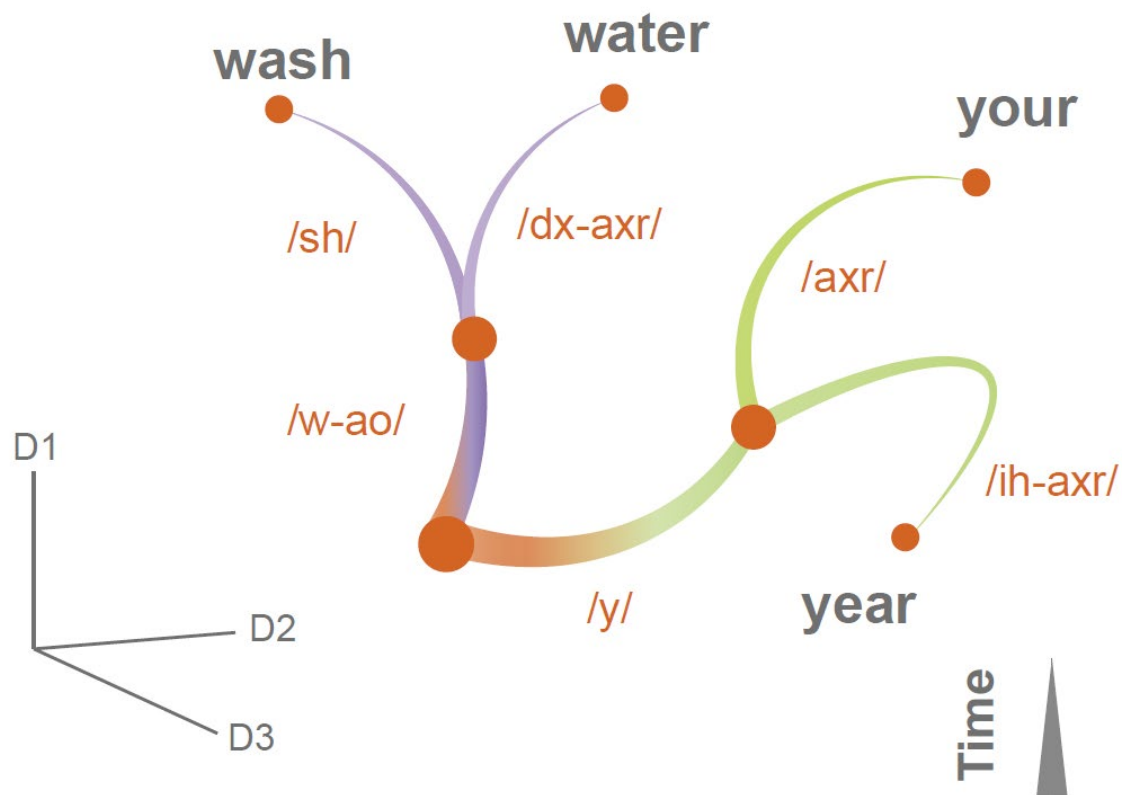
- ✓ The advantage of disentangled representation of order structure is that it enables the brain to **flexibly and rapidly process temporal sequences via conjunctive coding**, i.e, the brain can easily generate various temporal sequences by combining the temporal order with different contents.



- Understanding how the ordinal information is acquired and stored in the brain is crucial for us to understand how temporal sequences are processed in cognitive functions

The Neural Computing Mechanism

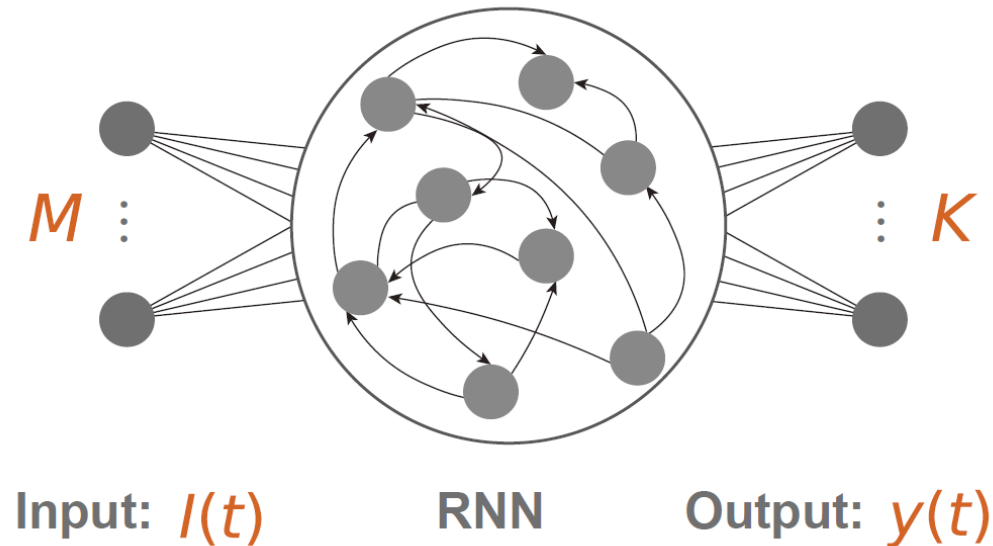
- ✓ The representations of temporal orders of related sequences display a **tree-like structure**.



- ✓ How recurrent neural circuits learn to represent the abstract **low-dimensional** order structure of temporal sequences?
- ✓ How the disentangled representation of order structure facilitates the processing of temporal sequences?

The Neural Circuits Model

- How biologically plausible recurrent neural circuits learn to represent the tree-structured orders of temporal sequences.



Input: conveying the information of temporal sequences.

Recurrent neural network: storing the ordinal structure of temporal sequences.

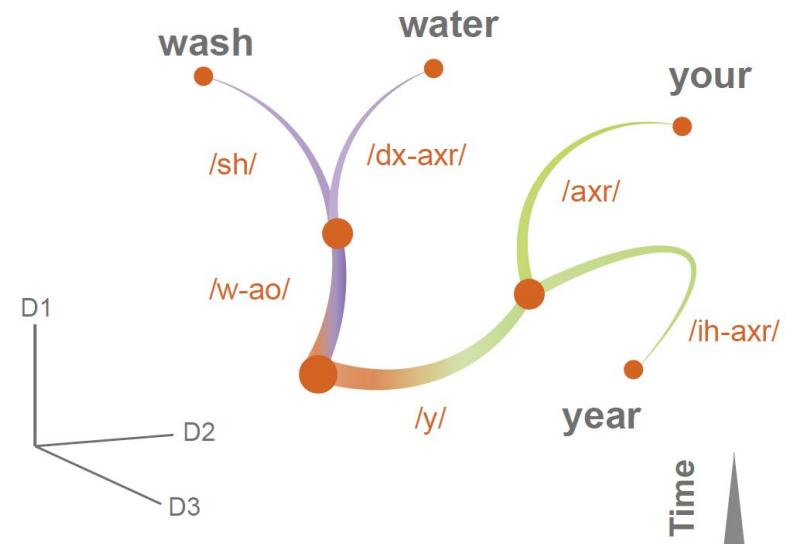
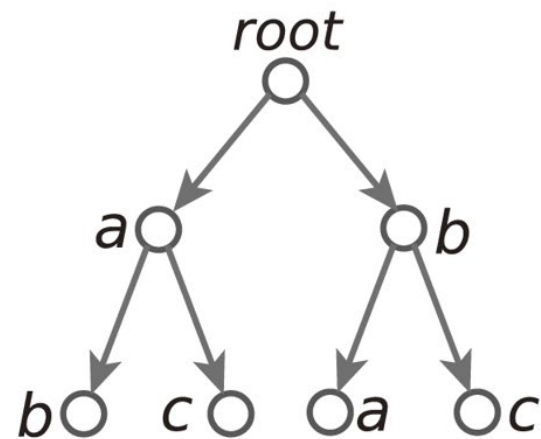
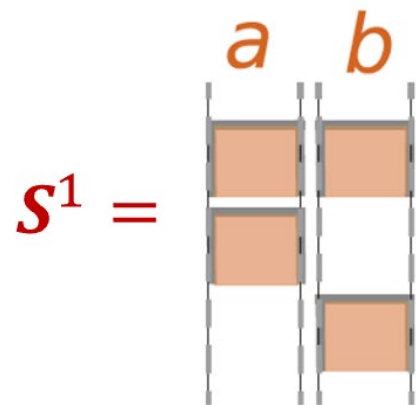
Output: discriminating temporal sequences.

The Training Protocol

□ A 4-class Synthetic Task

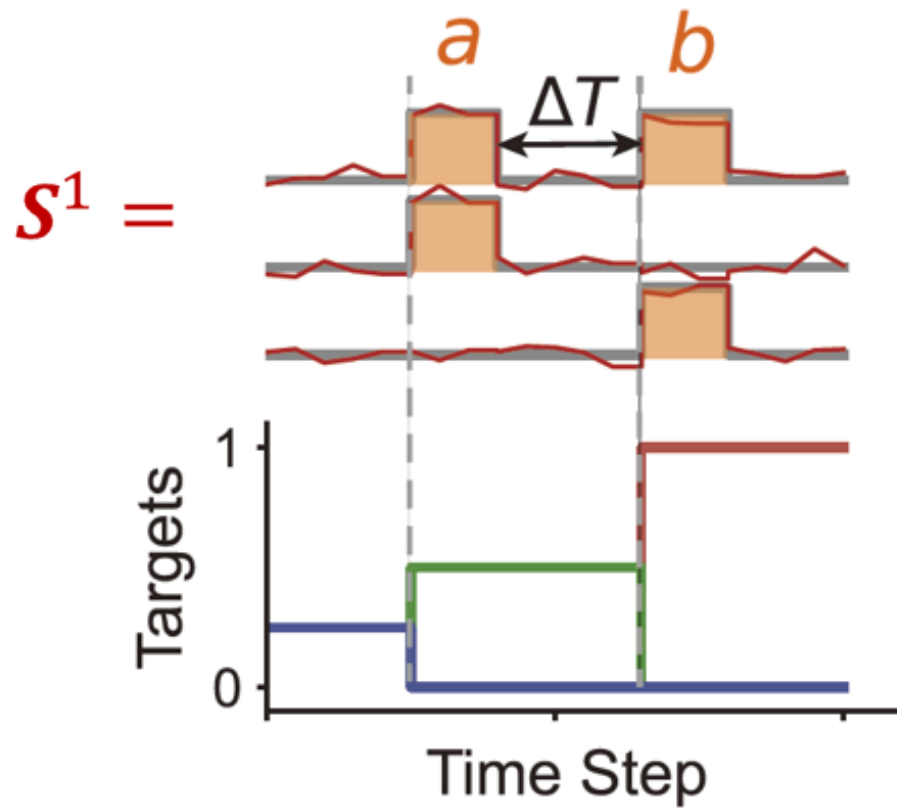
✓ The temporal sequences ($\mathbf{s}^k$, for $k = 1, \dots, 4$) are constructed by combining two of three primitive items ($\mathbf{a}, \mathbf{b}, \mathbf{c}$) with varied orders, where $\mathbf{a} = [1, 1, 0]^T$, $\mathbf{b} = [1, 0, 1]^T$, $\mathbf{c} = [0, 1, 1]^T$.

✓ $\mathbf{s}^1 = \{\mathbf{a}, \mathbf{b}\}$; $\mathbf{s}^2 = \{\mathbf{a}, \mathbf{c}\}$; $\mathbf{s}^3 = \{\mathbf{b}, \mathbf{a}\}$; $\mathbf{s}^4 = \{\mathbf{b}, \mathbf{c}\}$;



the temporal orders of the four temporal sequences form a two-layer tree structure

The Training Protocol



$$S^1 = \{a, b\}; S^2 = \{a, c\};$$

$$S^3 = \{b, a\}; S^4 = \{b, c\};$$

✓ Training data augmentation:

- via varying the duration between neighboring items in a temporal sequence (adding noisy signals).
- the model needs to realize invariant recognition with respect to the warping of a temporal sequence, which forces the RNN to extract the abstract ordinal structure of the temporal sequence.

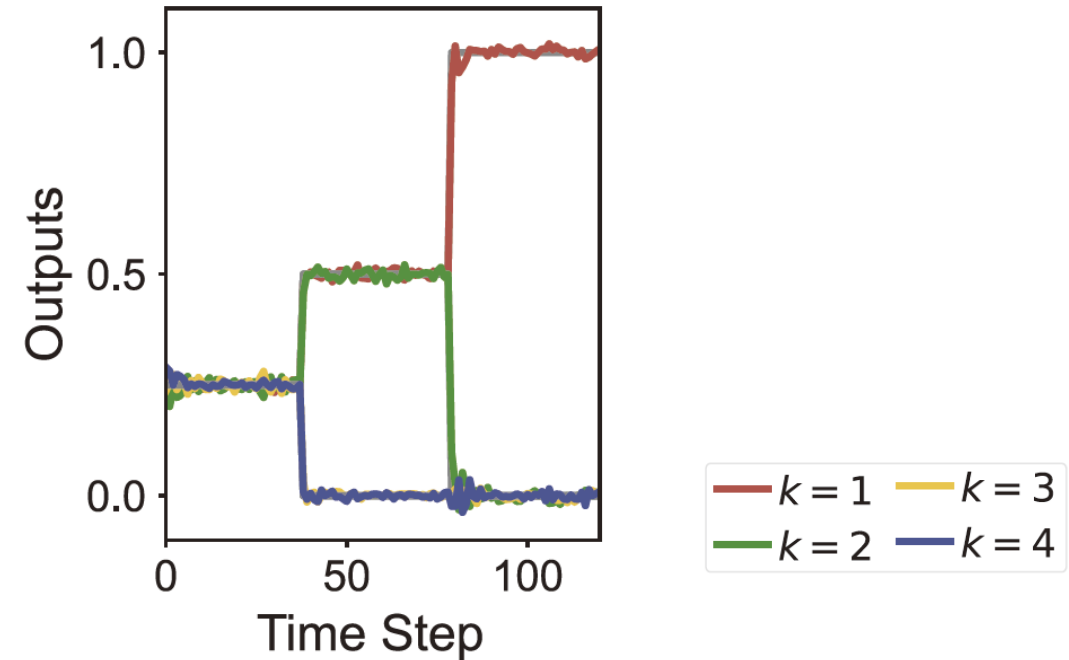
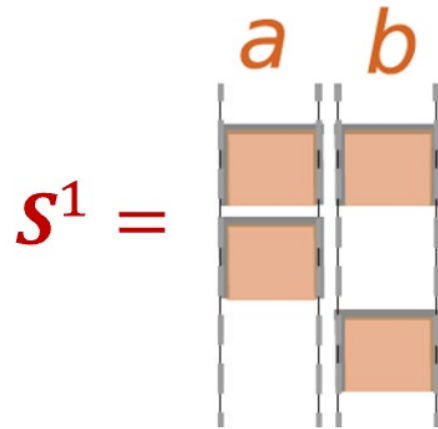
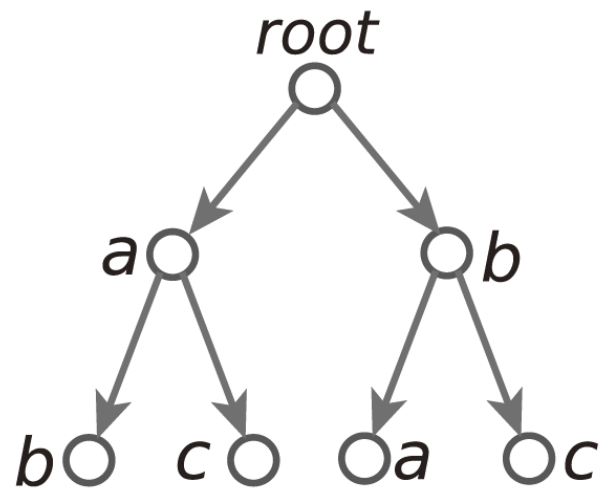
✓ Target function in response to temporal sequence encoded:

- Increases step by step as items in a temporal sequence unfold one by one.
- Since not all temporal sequences are differentiable by early items, neuronal responses should be low initially, and their values increase along with the exposure of followed items.

BPTT

Model Behavior

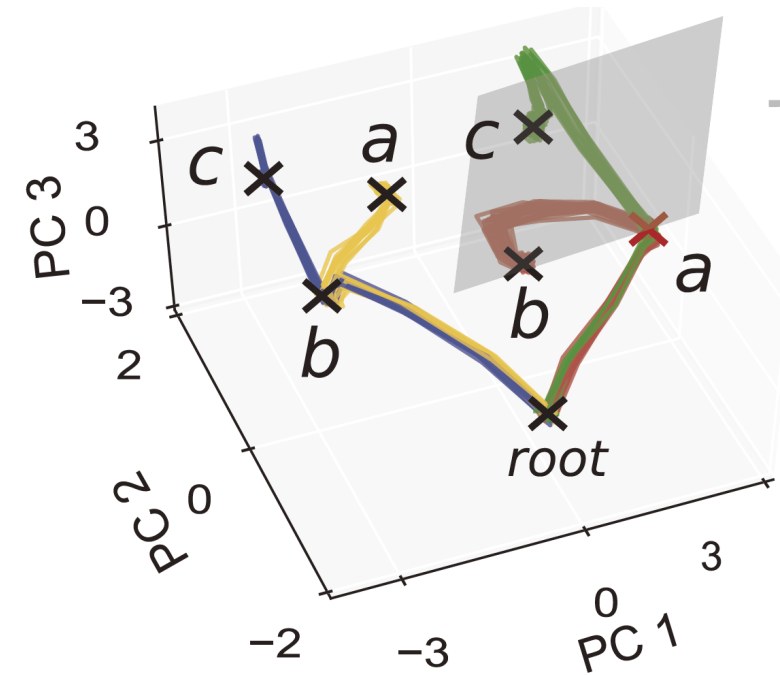
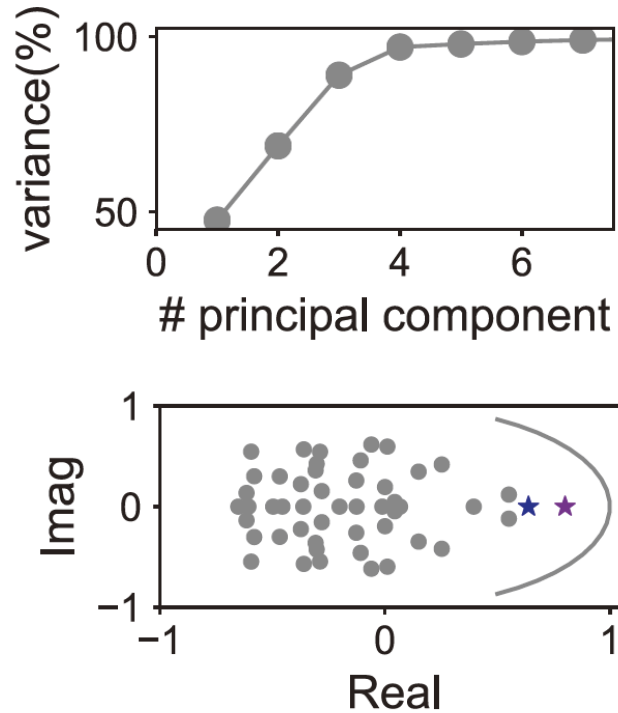
- ✓ Discriminating the temporal sequence in a coarse-to-fine manner.



$$S^1 = \{a, b\}; S^2 = \{a, c\}; S^3 = \{b, a\}; S^4 = \{b, c\};$$

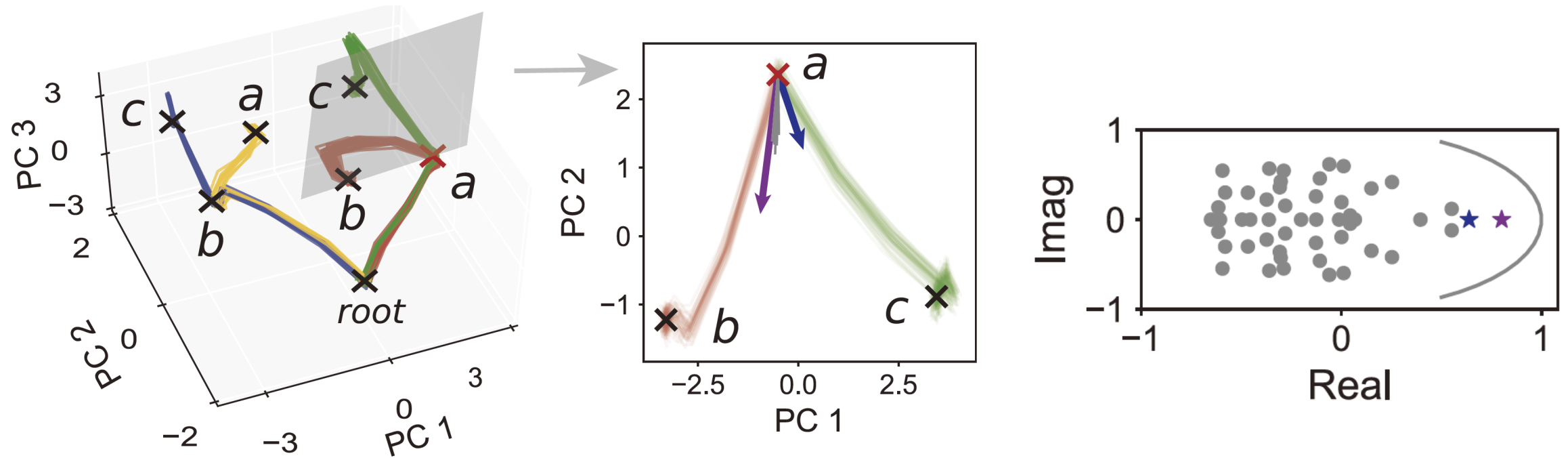
Visualizing the Learned Tree-structured Attractor Dynamics

- ✓ Tree-structured attractor states & attractor dynamics



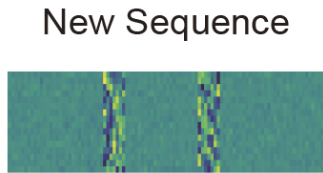
- By projecting network activities onto the top 3 PCs, the trajectories of the network state in responses to the four temporal sequences are obtained, which display a tree structure, agreeing with the ordinal structure of the four temporal sequences.

Visualizing the Learned Tree-structured Attractor Dynamics

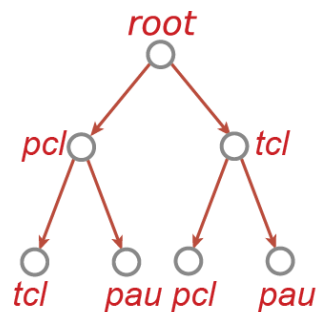


Disentangled Representation Facilitates Transfer Learning

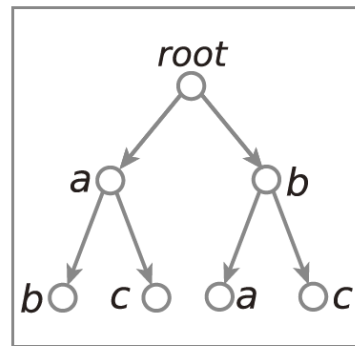
- The key advantage of disentangled representation of order structure is that it allows flexible generation to new temporal sequences via conjunctive code.
- Whether pre-trained RNN helps the model to speed up the acquisition of new temporal sequences having the same or partial ordinal structure, achieving the so-called transfer learning.



Ordinal Structure



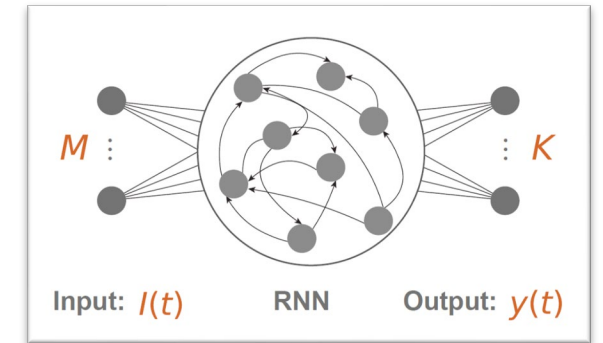
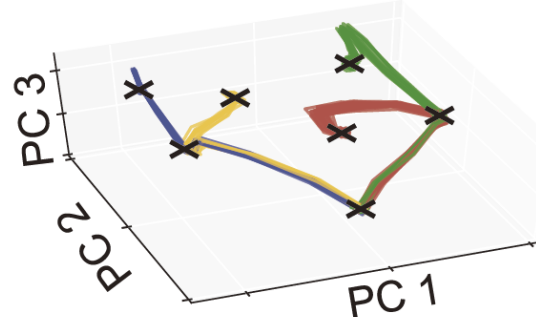
Template



Learned Dynamics

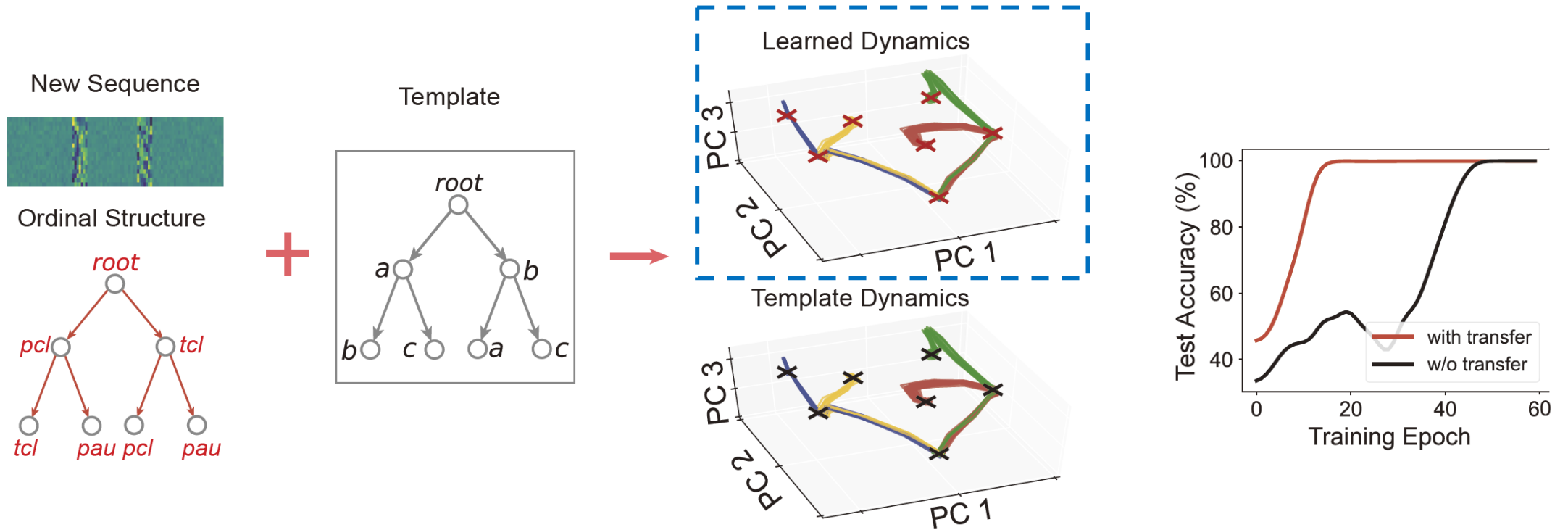
?

Template Dynamics



Via freezing the recurrent connections in the RNN, while only learning the feedforward connections from the input layer to the RNN and from the RNN to the output layer, when a new temporal sequence processing task arises.

Disentangled Representation Facilitates Transfer Learning



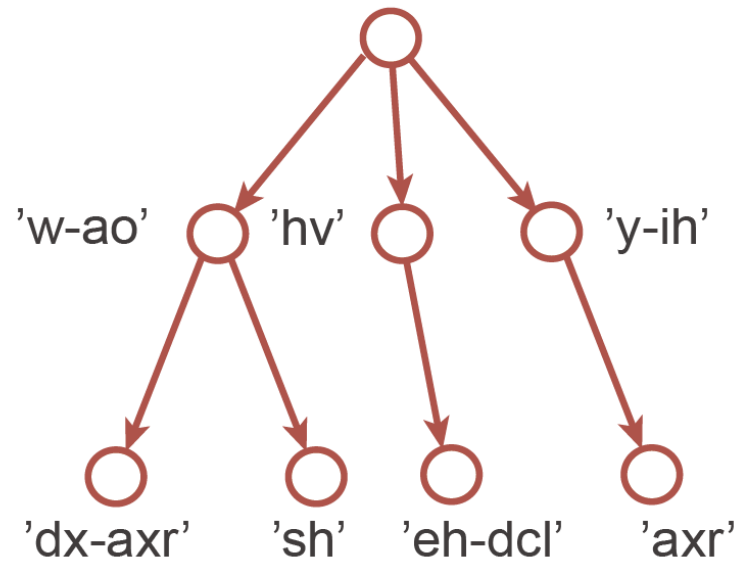
- After transfer learning, the RNN retains the tree-structured representation trajectories for the learned temporal sequences, indicating that the model adjusts the feedforward connections to match the tree-structured attractors of the RNN;
- The speed of transfer learning is accelerated significantly compared to that without freezing the RNN.

Robust Temporal Sequence Discrimination using Ordinal Structure

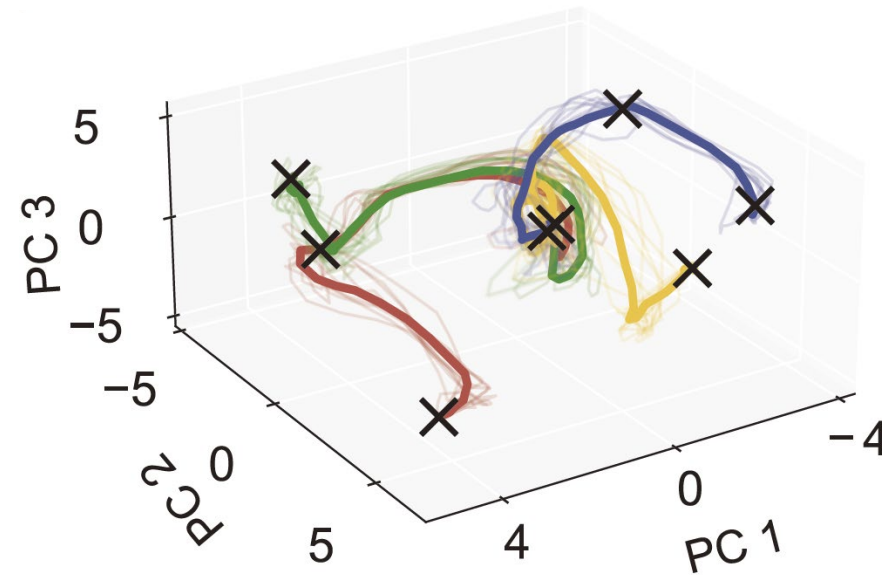
- A hallmark of temporal information processing in the brain is that the neural system can recognize temporally stretched or compressed signals, known as temporal warping, as the same pattern.
- **How neural circuits in the brain achieves this goal?**
- **Hypothesis:** storing ordinal structure as tree-structured attractor dynamics in recurrent neural circuits provides a feasible solution.

Robust Temporal Sequence Discrimination using Ordinal Structure

- **Key-words spotting:** A four spoken words discrimination task



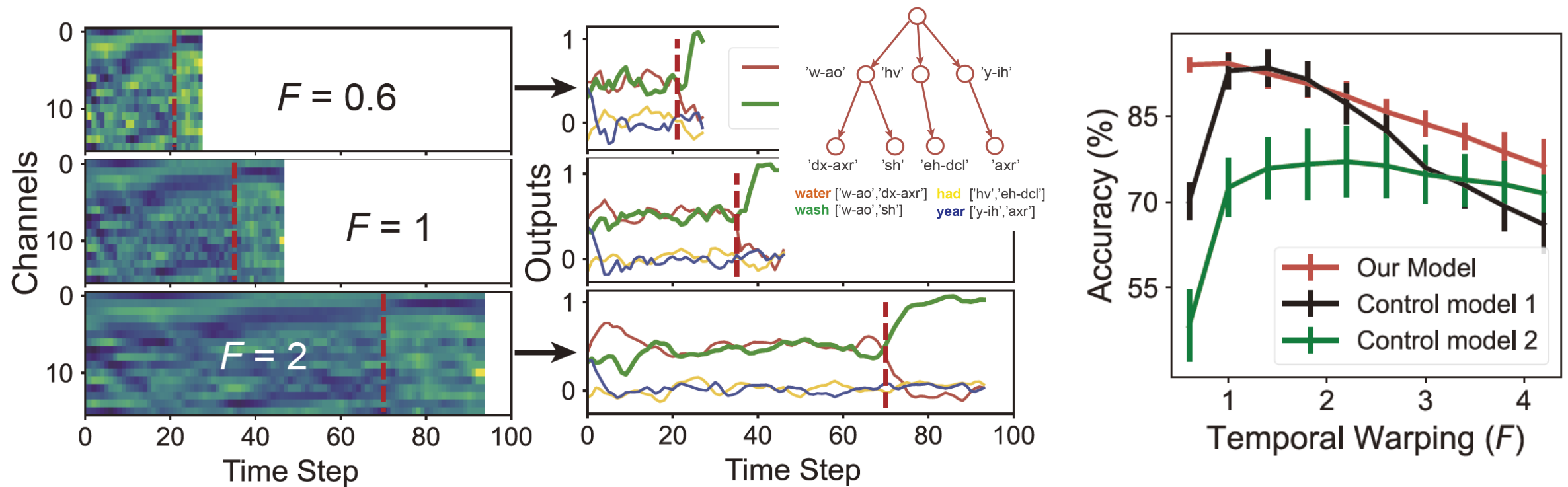
water ['w-ao', 'dx-axr'] **had** ['hv', 'eh-dcl']
wash ['w-ao', 'sh'] **year** ['y-ih', 'axr']



- The ordinal structure underlying the four spoken words is extracted.

Robust Temporal Sequence Discrimination using Ordinal Structure

In testing, we present temporal sequences of four words without inserting separation intervals, rather we stretch or compress the original temporal sequences to reflect the temporal warping effect in practice



F to denote the warping effect, with $F = 1$ representing no warping, $F > 1$ stretching, and $F < 1$ compressing.

Projects

1. Serial Dependence Effects in Large Language Model
2. Neural mechanisms of relational learning
3. The world model